

v-TCM: Vertical-aware Transformer Click Model for Web Search

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ABSTRACT

Understanding¹ and predicting user click behavior on a web search engine results page is critical for online advertising and recommendation engines. The click prediction results can be further used to estimate the relevance of the search engine results. Sequence-to-sequence models are effective for predicting user click behavior since they consider contextual information in the input sequence, and transformers are the recent state of the art in sequence learning. A novel transformer model is proposed in this paper for click prediction and relevance estimation that learns additionally from the vertical information, apart from the query and search engine results that are the inputs for the traditional click models. Vertical information is the representation style of a document in the search engine results page. The proposed vertical-aware transformer click model (v-TCM) takes into account both the position bias and the vertical bias of the search engine results. From our experiments, it is proved that v-TCM outperforms the existing click models for the click prediction and relevance estimation tasks.

CCS CONCEPTS

• **Information systems**→Information retrieval→Evaluation of retrieval results • **Computing methodologies**→Machine learning

KEYWORDS

Click prediction, relevance prediction, transformer, vertical information, web search

1 INTRODUCTION

Click models are the predictive models that are used for predicting user clicks over a browser's results page known as the Search Engine Result's Page (SERP). Earlier, most of the click models were based on Probabilistic Graphical Models (PGM) which is a mathematical way to encode the probability-based relationship

between different random variables in the form of a graph. Click models like dynamic bayesian model (DBN) [1], user browsing model (UBM) [2] and dependent click model (DCM) [3] are PGM based. Nowadays, most of the latest click models are based on deep neural networks. Click models like Neural Click Model (NCM) [4] and Click Sequence Model (CSM) [5] are some of the latest models based on the neural network framework that have outperformed the PGM-based models for click prediction [6].

The traditional click models have the query and the search engine results as their inputs. Most of the click models map the relationship between examinations and clicks, evaluating future clicks based on the past observations. There are relatively few works that have explored the incorporation of vertical information of SERP as one of the inputs to the click model. The result page for a search engine contains heterogeneous results federated from verticals of different types such as multimedia, application and text verticals (examples: Image, Video or News), along with the ordinary or organic results, as illustrated in Fig. 1.

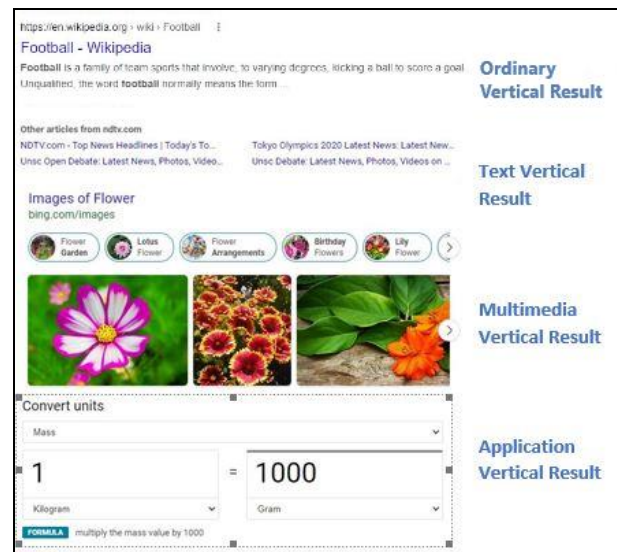


Figure 1: Different types of verticals in SERP

These vertical results seriously influence the user behavior during browsing. Users tend to get attracted by the vertical results and their visual presentation, that may increase the examination probability of nearby results [7]. Examples of some previous click models that have made use of verticals are the vertical-aware click

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model in [8] and the federated click model in [9]. Chen *et al.* [9] were one of the first to model user behavior using vertical results. They found that users are more likely to examine the vertical results and the ordinary web documents nearby. The intuitiveness test by Chuklin *et al.* in [10] established that the document placed near an attractive vertical will have more relevance score. Their work highlights the significance of the placement of verticals for relevance estimation of the search engine results. The vertical-aware model in [8] proposed improvements by incorporating the vertical results, but the relationship between the vertical score and click probability is hard-coded and that may restrain system performance. To shed light on the above-mentioned issue, we have proposed a novel click model that incorporates the vertical/non-Web-page results of the SERP into the input embedding layer of a multi-head attention transformer model [11]. Transformers are the latest state of the art in sequence-to-sequence learning that are rapidly replacing Long Short-Term Memory (LSTM) [12] in the encoder and decoder modules. Some recent works have started adapting transformers for constructing click models. Li *et al.* (2020) proposed the InterHAt model [13] that uses a transformer with multi-head self-attention for feature learning for the task of click prediction. The distinguishing feature of this model is that it uses attention aggregation to capture high-order feature interactions. Our contribution is new from the perspective that we incorporate the vertical information type of each document in SERP (vertical bias) as an additional input to the encoder of a multi-head self-attention based transformer, apart from the query and the ranked search engine results (position bias) which is the conventional input of click prediction models. The main contributions of our paper are summarized as follows.

- A novel vertical-aware transformer click model called v-TCM is proposed which takes the vertical bias into consideration. The new model is able to incorporate both the position and vertical biases into a calculable framework.

- Transformers with multi-head self-attention constitute the state of the art in sequence learning, and this model is used for learning the position and vertical biases from user logs.
- The model achieves significantly better performance in both click prediction and relevance estimation as compared to existing click models.

The organization of the rest of this paper is as follows. Section 2 describes the proposed model, Section 3 describes the experimental setup and results, and Section 4 draws the final conclusions.

2 VERTICAL-AWARE TRANSFORMER CLICK MODEL

In this section, we describe the architecture of the proposed vertical-aware Transformer Click Model. Click modeling can be interpreted as a sequence modeling task. The query and the search engine results define the input sequence, and the prediction of whether the user will click on a search result or not is recorded for the 10 results displayed on the search page, and this defines the output sequence. Recurrent neural networks, specifically LSTMs, have been used for sequence-to-sequence modeling and prediction [12, 14]. Neural networks are found to capture context well using attention mechanism [15], and that improves the click prediction and relevance estimation results, as proved by researchers in [16]. However, since its inception in 2017 [11], transformers have overtaken LSTMs in most of the sequence modeling tasks. The primary advantage is the parallel computing and multi-head self-attention incorporated in the transformer architecture. In our work we propose to use the transformer for click prediction. We feed the query, position bias (ranked search engine results) and vertical bias as input to the transformer encoder, along with the target clicks. Fig. 2 shows the architecture of the proposed vertical-aware transformer click model that we call as v-TCM.

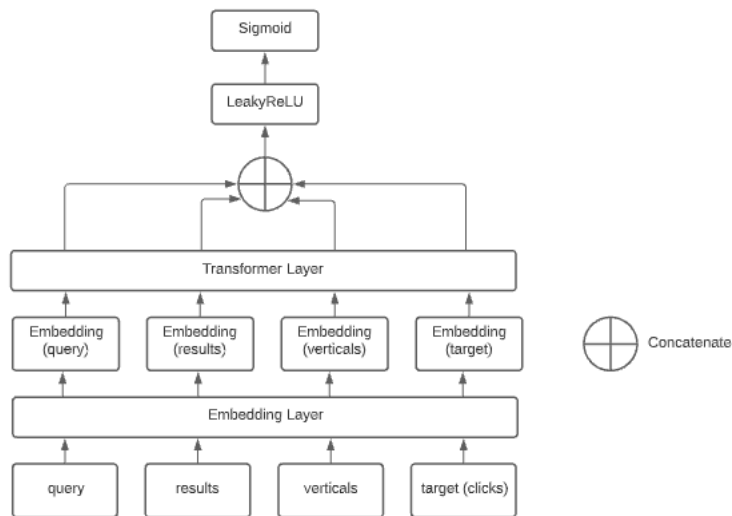


Figure 2: The architecture of the proposed vertical-aware Transformer Click Model (v-TCM)

The transformer PyTorch embedding layer is used to embed all the input features into distributed fixed-size 192-dimensional vectors. The input features are query q , search engine result d_t , and the vertical type v_t of each result, where $t=1,2,\dots,10$. We concatenate all the input embeddings and that is given as input to the transformer encoder, along with the embedded target click sequence, as shown in Fig. 2. Our multi-head transformer architecture used four heads whose outcomes are concatenated as shown in Fig. 2. To avoid overfitting, we make use of dropout and Leaky ReLU in both self-attention and feed forward network sub-layers. We then use three fully connected layers to learn the interaction among the dense features. Since the problem is related to whether the user will click on a search result or not, therefore it is a binary classification problem, and thus we use the sigmoid function in the output unit. To train the model, the loss function used is the cross-entropy loss function.

3 EXPERIMENTAL SETUP AND RESULTS

Through our experiments, we aim to answer the following research question: - does the transformer-based click model perform better in terms of click prediction and relevance estimation, and how incorporating the verticals into the click models affects the overall performance. In this section, we first outline the experimental setup. The results obtained are then analyzed with respect to the click prediction accuracy and relevance estimation.

3.1 Dataset

We conducted our experiments on an open log-based session dataset named TianGong-ST [18]. The details are shown in Table 1. The SERP data is refined from the 18-day user log recorded by search engine Sogou.com. The dataset is split at the source into train, test and validation sets. The model is trained on the training set. The hyperparameters were set based on the model's performance on the validation set. Finally for calculating the performance and evaluation metrics, the testing set was used. As noted from Table 1, there are 4491 unique verticals in this dataset that are assigned unique IDs.

Table 1: Dataset statistics

Dataset	TianGong-ST
#sessions	147,155
#unique queries	40,596
#unique doc	263,384
#unique verticals	4491
search engine	Sogou.com
collection	newly crawled in Dec, 2018 (Updated Aug, 2020)

3.2 Performance metrics

We evaluate the performance of v-TCM for click prediction and relevance estimation. For evaluating the click prediction results,

we compute the perplexity (PPL) on the test set. The definition of click perplexity (PPL) at the rank r is given by

$$\text{PPL}(\mathbf{M}) = 2^{\frac{1}{|S|} \sum_{s \in S} \log_2 P_{\mathbf{M}}(\mathbf{C}_r = \mathbf{c}_r^{(s)})} \quad (1)$$

where, $\text{PPL}(\mathbf{M})$ denotes the total perplexity of the model $\mathbf{M}$. $P_{\mathbf{M}}(\mathbf{C}_r = \mathbf{c}_r^{(s)})$ denotes the probability of click event $\mathbf{c}_r^{(s)}$ having rank r in a query session s . Here, S is the total number of sessions. The final perplexity of the model $\mathbf{M}$ is calculated by averaging the perplexities over all positions or ranks. Lower the value of perplexity, the higher will be the quality of the model. For relevance estimation, we use the relevance scores estimated by the click models to re-rank the document list, and calculate the normalized discounted cumulative gain (NDCG) according to the human annotation provided in the TianGong-ST dataset. A higher value of NDCG denotes better relevance estimation by the model.

3.3 Baseline models

We compare the performance of our transformer click prediction model v-TCM with state-of-the-art click models like UBM, DCM, DBN, NCM and CSM. While UBM, DCM, DBN are probabilistic graphical models, NCM and CSM are neural models based on LSTM. We implemented these models in accordance with the guidelines in their original papers [1-5]. We have also included the transformer model (without vertical information) proposed recently for Alibaba recommendation machines [17].

3.4 Hyperparameter settings

For training the proposed models, we use a batch size of 64 and the Adam optimizer. Both the hidden layer size and the embedding dimension is fixed to 64. The initial learning rate and dropout rate is set to 0.001 and 0.5, respectively. For each query, the top 10 documents are considered. Weight decay parameter (λ) is set to 10^{-5} to avoid model overfitting. After every 5 iterations we check the model's validation performance and if it is less than the current validation click perplexity, we update it. The entire coding is done in Pytorch using Google Colab notebook, and all the deep models are trained on the Nvidia K80 GPU.

3.5 Results and discussion

In this sub-section, we analyze the results obtained for the proposed and baseline models, which are summarized in Table 2.

Table 2: Click prediction results

Model	Perplexity	NDCG@10
UBM [2]	1.1793	0.8395
DCM [3]	1.1816	0.8238
DBN [1]	1.1846	0.8346
NCM [4]	1.1758	0.8423
CSM [5]	1.1724	0.8567
Transformer [17]	1.1782	0.8598
v-TCM (Ours)	1.1704	0.8622

The baseline models do not include the vertical information as input features. Table 2 shows the average perplexity score and the NDCG@10 score for v-TCM and the state of the art click models UBM, DCM, DBN, NCM, CSM, and transformer without vertical information. As observed, v-TCM achieves the best scores with the highest NDCG@10 value (0.8622) and the lowest perplexity value (1.1704). Fig. 3 showcases the perplexity scores of the three best performing models in Table 2, namely, NCM, CSM, and v-TCM, over different search result positions ranging from 1-10. As we can infer from Table 2 and Fig. 3, v-TCM performs better as compared to the baseline models due to the lower perplexity values at all search result positions, especially for the top ranked results (ranks 1 to 4).

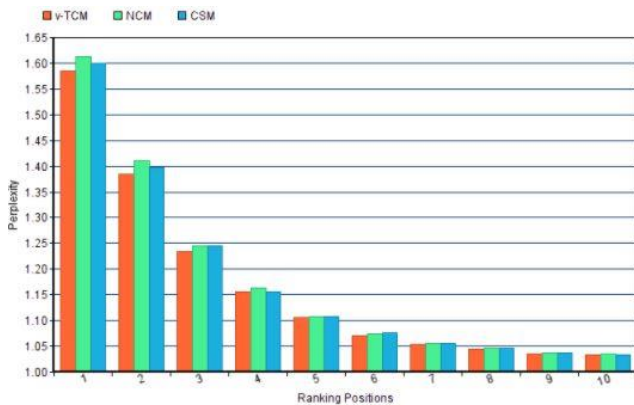


Figure 3: Perplexity scores of NCM, CSM, v-TCM click models for different positions in the SERP

Thus, we can infer that incorporating the verticals into click model improves the performance of the model. The reason is that since the model is taking both position bias and the vertical information into account, this interprets user click behavior better in terms of both perplexity and NDCG@10. Also, the architecture of the multi-head self-attention transformer model improved the effectiveness of our approach.

4 CONCLUSIONS

In this paper, we propose a novel vertical-aware transformer click model (v-TCM) in which we additionally incorporate vertical information as an input to the transformer encoder, apart from the query and search engine results. Experiments are conducted on a public web log session dataset. We find that: 1) transformer model is performing better in terms of perplexity i.e. click prediction, and NDCG@10 i.e. relevance estimation, as compared to the baseline models. 2) incorporating the vertical as an additional input proves fruitful for our cause since it significantly improves the performance, as compared to the baseline models like CSM and NCM that do not take the verticals into account as per their model architecture. More user behavior information must be exploited in the future for improved session and click modeling, like incorporating the vision bias and mouse movement as additional input features in the proposed click model.

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